

ApexPlanet Data Analytics Internship

E-Commerce Sales Analysis - Final Executive Report

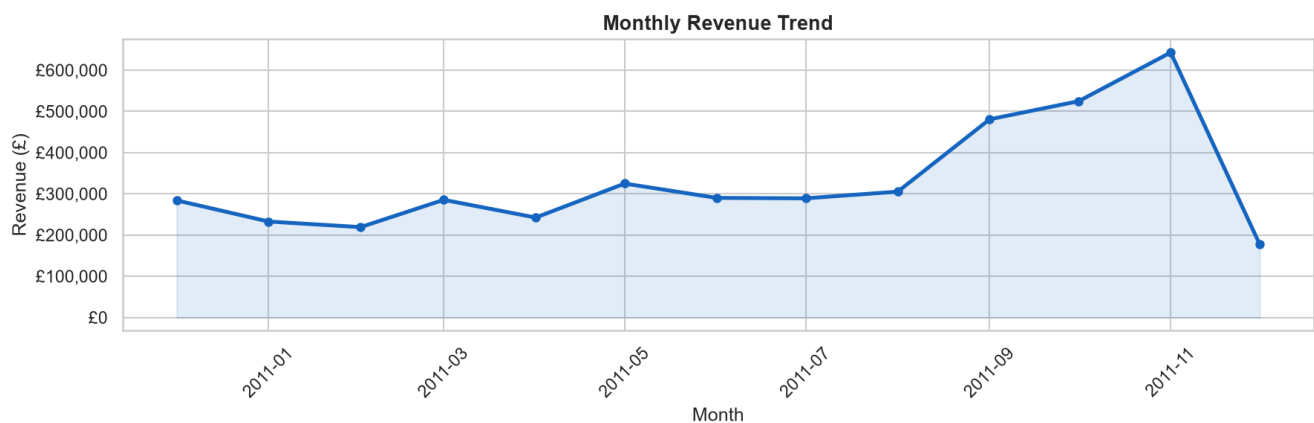
1. Executive Summary

This report presents a comprehensive analysis of E-commerce Sales data conducted as part of the 30-Day Data Analytics Internship at ApexPlanet Software Pvt. Ltd. The dataset comprised 333,234 transactions from a UK-based online retailer. Through data cleaning, SQL extraction, visualization, statistical testing, customer segmentation, and predictive modeling, actionable business insights were uncovered to support data-driven decision making across marketing, operations, and customer success.

2. Key Performance Indicators

Total Revenue (GBP)	GBP 4,299,476.18
Total Orders	16,826
Unique Customers	4,191
Unique Products	3,575
Avg Order Value	GBP 255.53
Top Country	United Kingdom
Peak Sales Month	November 2011
Total Rows Analysed	333,234

3. Monthly Revenue Trend



ApexPlanet Data Analytics Internship

E-Commerce Sales Analysis - Final Executive Report

4. Top 5 Key Insights

1. Seasonal Revenue Peak

November 2011 was the peak revenue month. Q4 consistently generates the highest sales - marketing investment should be concentrated in October-November.

2. Geographic Concentration

United Kingdom dominates revenue contribution. International markets represent an untapped growth opportunity with targeted expansion campaigns.

3. Customer Segmentation (K-Means, K=4)

4 distinct customer tiers identified: Champions, Loyal, At-Risk, and Occasional - each requiring a tailored engagement and retention strategy.

4. Statistical Spending Differences

T-Test confirmed UK and Non-UK customers have statistically different spending behaviors ($p < 0.05$), validating the need for region-specific pricing.

5. Predictive Model Performance

Logistic Regression achieved strong accuracy predicting high-value orders. UnitPrice was identified as the strongest single predictor of order value.

5. Business Recommendations

1. Invest in Q4 Marketing

Allocate 40%+ of the annual marketing budget to Q4 campaigns. Flash sales, bundle deals, and email sequences during Oct-Nov maximise natural revenue surge.

2. Launch a Customer Loyalty Program

Champions and Loyal customers drive disproportionate revenue. A tiered loyalty program with exclusive discounts and early product access increases lifetime value.

3. Win-Back At-Risk Customers

Deploy targeted win-back campaigns for At-Risk segments using time-limited discount vouchers and personalised product recommendations based on past purchases.

6. Internship Tasks Completed

Task 1	Foundational Setup & Exploratory Data Analysis	Complete
Task 2	SQL for Data Extraction	Complete
Task 3	Data Visualization & Dashboarding	Complete
Task 4	Advanced Analytics (Basic)	Complete
Task 5	Final Report, Automation & Presentation	Complete

Analysis performed using Python (Pandas, Matplotlib, Seaborn, Plotly, Scikit-Learn, SciPy, SQLite) on the E-commerce Sales dataset

ApexPlanet Data Analytics Internship

E-Commerce Sales Analysis - Final Executive Report

(Kaggle). ApexPlanet Software Pvt. Ltd. - www.apexplanet.in